

Preliminary Project Report

FHSS Demodulation on Hardware

Camille Ferrell & Liam McCarthy

Project Description:

When signals are broadcast over a particular frequency, they become vulnerable to jamming and eavesdropping, which is a major security risk for any information that needs to remain private. One solution to this problem is to skip, or hop, between frequencies at a rate fast enough that another party cannot keep up. By having a transmitter and receiver hopping according to the same sequence, the sender and listener can both make sense of the signal while thwarting any adversaries' attempts to interfere with the message. [1]

For our project, our goal was to implement this frequency-hopping spread spectrum behavior in both hardware and software on an FPGA board to examine the differences in speed and reliability of the processor state and programmable logic implementations.

Progress Report:

So far, we have implemented the software modulation and demodulation for our FHSS in Python. This implementation is working, and we demonstrated the capability in update 1.

We have also implemented the hardware demodulation, and attempted to run it in our Jupyter notebook. However, the program stalled and did not provide outputs, but we think we have identified the problem, which is that we did not enable `AUTO_RESTART` for our `axi_stream`. We plan to implement this change next week, which will hopefully resolve the problem; otherwise, we will have to look for alternative solutions or think about changing our block diagram.

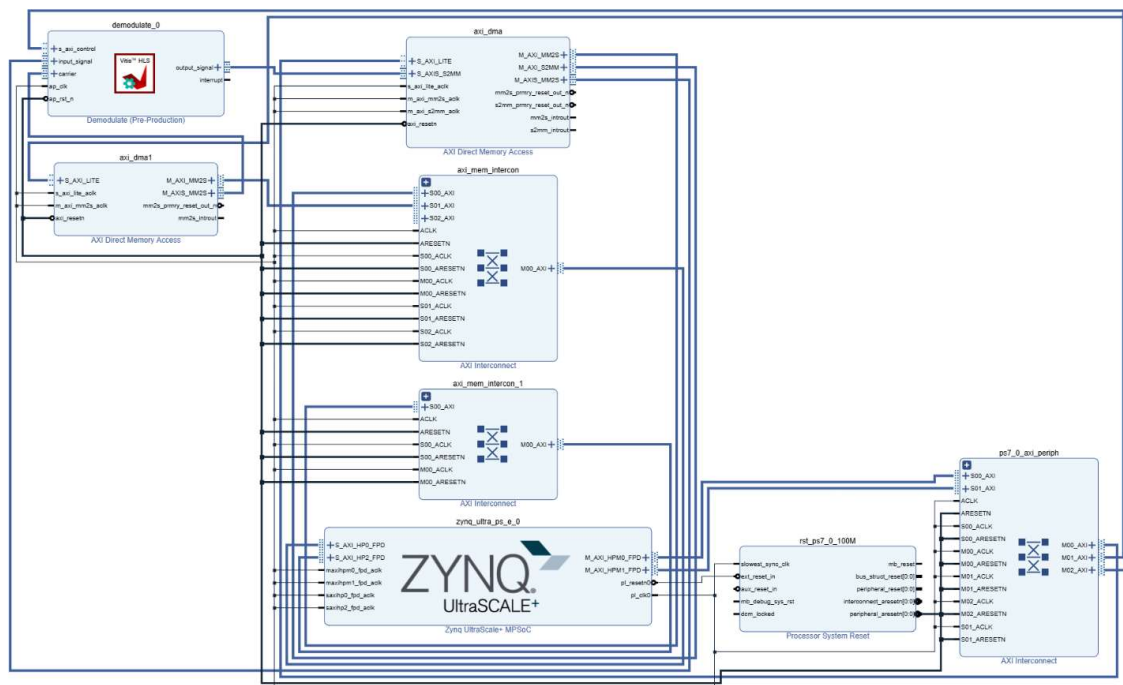


Figure 1: Overlay block diagram

Division of Work:

	Work
Combined Effort	• Vivado block design and overlay • Jupyter notebook implementation
Camille	• Modulation/demodulation of .wav

	files • Vitis code for hardware implementation
Liam	• Packetization of .wav files and .bin conversion • Pseudo random FHSS implementation

Sources

[1]

https://cs.stanford.edu/people/eroberts/courses/soco/projects/2003-04/wireless-computing/int_ss.shtml